Temperature Controller – Maintenance Practices

1. Adjustment/Test

A. Function Test Temperature Controller

Equipment required:

- DC supply – Adjustable 15 – 30 volts
- Multimeter
- Decade Resistance Box 0 to 400 ohms in 0.1 ohm steps (10 ma rating)

1) Check temperature operating point:

a) Ensure aircraft power is OFF.

b) Disconnect electrical connector from controller.

c) Connect multimeter (set to ohms range) between pins J and K of receptacle on controller.

d) Apply 28 Volt DC (positive to pin E, negative (ground) to pin F).

e) Connect decade resistance box between pins A and B. Set decade to 330 ohms.

f) Turn temperature adjustment control (R14) set screw on top cover of controller until relay K1 picks up (indicated when meter reads zero ohms).

g) Vary decade resistance to determine “dead band”. Check that the difference between the pickup and dropout resistance is 7 +2 or –3 ohms.

h) Set decade resistance to 300 ohms. Vary R14 to check that relay K1 picks up.

i) Set decade resistance to 370 ohms. Vary R14 to check that relay K1 picks up.

j) Reset resistor R14 to pick up relay K1 with decade resistance setting of 330 +5 or –4 ohms.

k) Adjust input voltage to 22 Volt DC. Check that relay K1 pickup and dropout points do not differ by more than 1.6 ohms from values obtained in Para 1.A.1).j).

l) Check that relay K1 does not chatter when decade resistance is varied, and that the “dead band” is within ± 0.5 ohm of value obtained in Para 1.A.1).g).

2) Fail-safe test:

a) Adjust input voltage to 28 volts, and set decade resistance to pick up relay K1.

b) Remove connection from pin A or pin B. Check that relay K1 immediately drops out (indicated by meter reading infinity).

c) Set decade resistance to 1 ohm greater than value required for relay K1 to pick up.

d) Apply ground at pin A (jumper pin A to pin F) and check that relay K1 drops out. Repeat by applying ground at pin B.

e) Disconnect meter, decade resistance and power supply. Reconnect electrical connector to temperature controller.

3) Continuity test:

With all power disconnected from controller, check pin to pin resistance as follows using multimeter:

a) Between connector pins J, K, I, C, L, N, H, D, M and G to all other pins. Check for infinity.

b) Pin A to pin E; 250 ohms ± 5%.

c) Pin A to pin F; 250 ohms ± 5%.

4) Zener diode test:

a) Apply 28 Volt DC (positive to pin E, ground to pin F) through a 0 to 0.5 ampere DC ammeter.

b) Check that ammeter indicates 175 to 250 milliamperes if R12 is 47 ohms and 248 to 350 milliamperes if R12 is 33 ohms.

Note In certain units a 33 ohm resistor is installed in location R12 instead of a 47 ohm resistor. When CR1 is changed, check that R12 is 47 ohms and replace if necessary.

c) Adjust input supply to 22 Volt DC. Check that ammeter current is 47 to 123 milliamperes if R12 is 47 ohms, and 66 to 176 milliamperes if R12 is 33 ohms.

d) Adjust input supply to 29 Volt DC. Check that ammeter current is 195 to 270 milliamperes if R12 is 47 ohms, and 279 to 388 milliamperes if R12 is 33 ohms.